

GROUP 38

MIS774 Assignment 1

ICE Entertainment Case Study

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BRIDGING THE GAP BETWEEN GLOBAL GROWTH AND FRAGMENTED DATA



Background:

ICE Entertainment has achieved massive growth across 500 global stores.

Challenge:

Highly siloed critical business data.

Informix system → sales and customers data

Oracle ERP → product costs and inventory

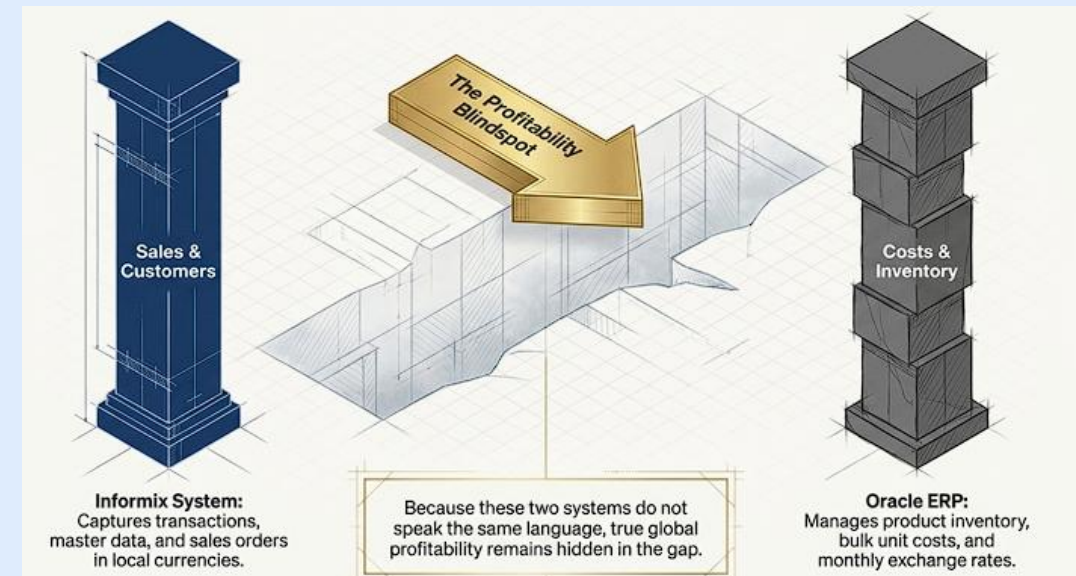
Pain Point:

Disconnected Systems → Invisibility of Global Profitability

→ Intuitive Decisions > Evidence-driven decisions

Solution:

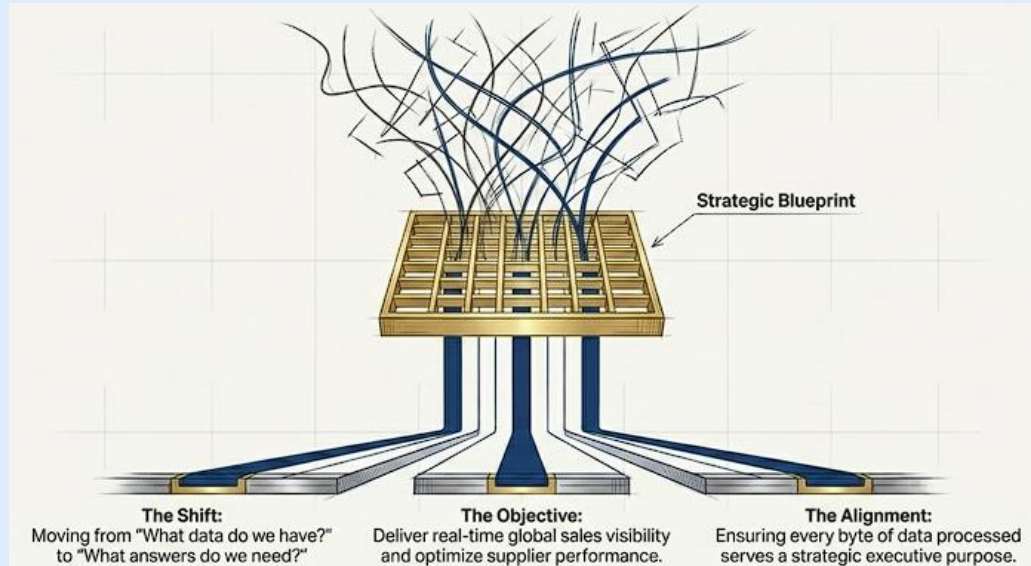
Develop an enterprise-wide dimensional model to integrate these systems, establishing a "single source of truth" for all management reporting.



IMPLEMENTING OUR STRATEGY WITH DATA STRATEGY CANVAS

1. Strategic Objectives - Single Source of Truth - Evidence-Based Decision Making - Global Sales Visibility - Purchasing Intelligence - Customer Loyalty Foundation	Strategic Framing		Strategic Enablement	
	2. Key Decisions & Use Cases: - Customer Profitability - Product Performance - Location Intelligence - Temporal Analysis - Market Segmentation	3. AI / Analytics Opportunities: - Predictive analytics for customer lifetime value and churn (loyalty program) - Anomaly detection in supplier performance - Seasonal demand forecasting for procurement - Recommendation engines for subscription packages - Automated margin alerts when streaming costs (Geo Broadcasting) erode profit	5. Data Requirements & Products: Required data: - Sales transactions (Informix) + product costs & currency rates (Oracle) - Customer, store, product, time, market hierarchies - Historical location (slowly-changing dimensions) Products to build: - Central Sales Fact Table (star schema) in the Gold Layer - Conformed dimensions - Data marts / semantic layer for the 5 board questions - Dashboards & reports + ad-hoc analytics layer	6. Consumers & Users: - CEO Rod, Sales Manager Pearce, Purchasing Head Milton, Board Analysts & CRM team - Future loyalty program team
	4. Value Measures & ROI/CBA: - ROI metrics : Faster global profit reports - CBA : Cost of building dimensional DW + ETL vs. revenue uplift		7. Technology & Systems: - Source systems: Informix (Sales) + Oracle (ERP) - Target: Modern analytical ecosystem (on-prem/cloud/hybrid) – Staging / Data Lake (Bronze) – Clean Silver Layer – Dimensional DW + Data Marts (Gold Layer) - ETL/ELT pipelines for currency conversion & SCDs - BI tools (dashboards, semantic layer, ad-hoc tools)	
Strategic Readiness				
8. Portfolio Management: - Directly supports the board-approved enterprise analytical solution initiative - Dependencies: data integration project between Informix & Oracle - Connects to: upcoming Loyalty Program launch and procurement transformation - No major competing programmes identified	9. Operating Model & Governance: - Business domains (Sales, Purchasing, CRM) own definitions - Central Data Team owns the dimensional model - Governance: Data quality rules, monthly currency-rate validation, SCD handling for market redefinitions - Sustainable model: automated ETL + monitoring, ethical use of customer data		10. Risks & Ethics: - Risks: Currency conversion errors → wrong margins - Privacy - Market segmentation changes without reprocessing history - Bias in age-group / loyalty segmentation - Mitigation: Use surrogate keys + slowly-changing dimensions (Type 2) - Monthly exchange-rate audit table - GDPR/CCPA-compliant customer data handling - Clear documentation of "location at time of sale" - Ethical AI guidelines for future predictive models	

ALIGNING TECHNICAL ARCHITECTURE WITH EXECUTIVE VISION



Data Architecture:

Implement an ETL pipeline (Pentaho) to extract and transform incompatible data from Informix and Oracle into a unified MySQL dimensional schema.

Value Proposition:

Standardize global reporting into USD and empower the executive team to make rapid, evidence-based strategic decisions.

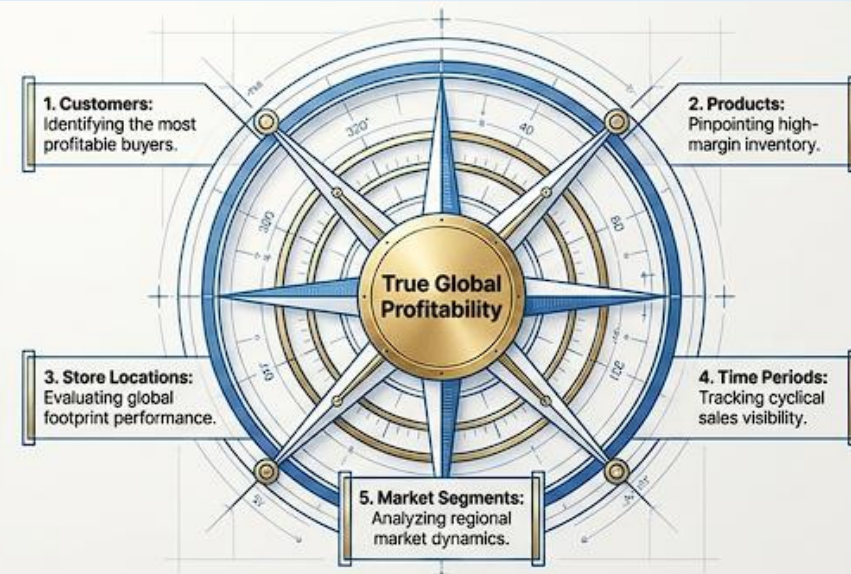
Strategic Objectives:

- Deliver real-time global sales visibility
- Optimize supplier performance and inventory management
- Design for a new customer loyalty program

Key Business Questions:

Determine true profitability across five core dimensions:

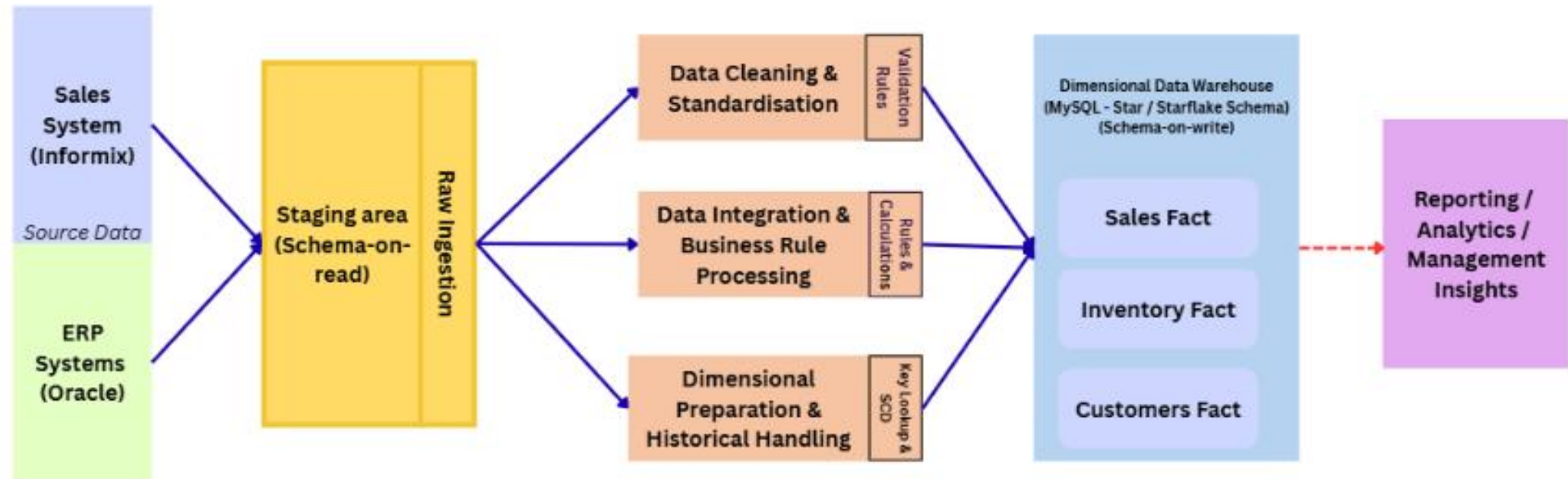
Customers, products, store locations, time periods, and market segments



By aligning the technical "how" with the C-suite "why," we illuminate these five distinct drivers of revenue.

DATA FLOW ARCHITECTURE

Data Flow Architecture

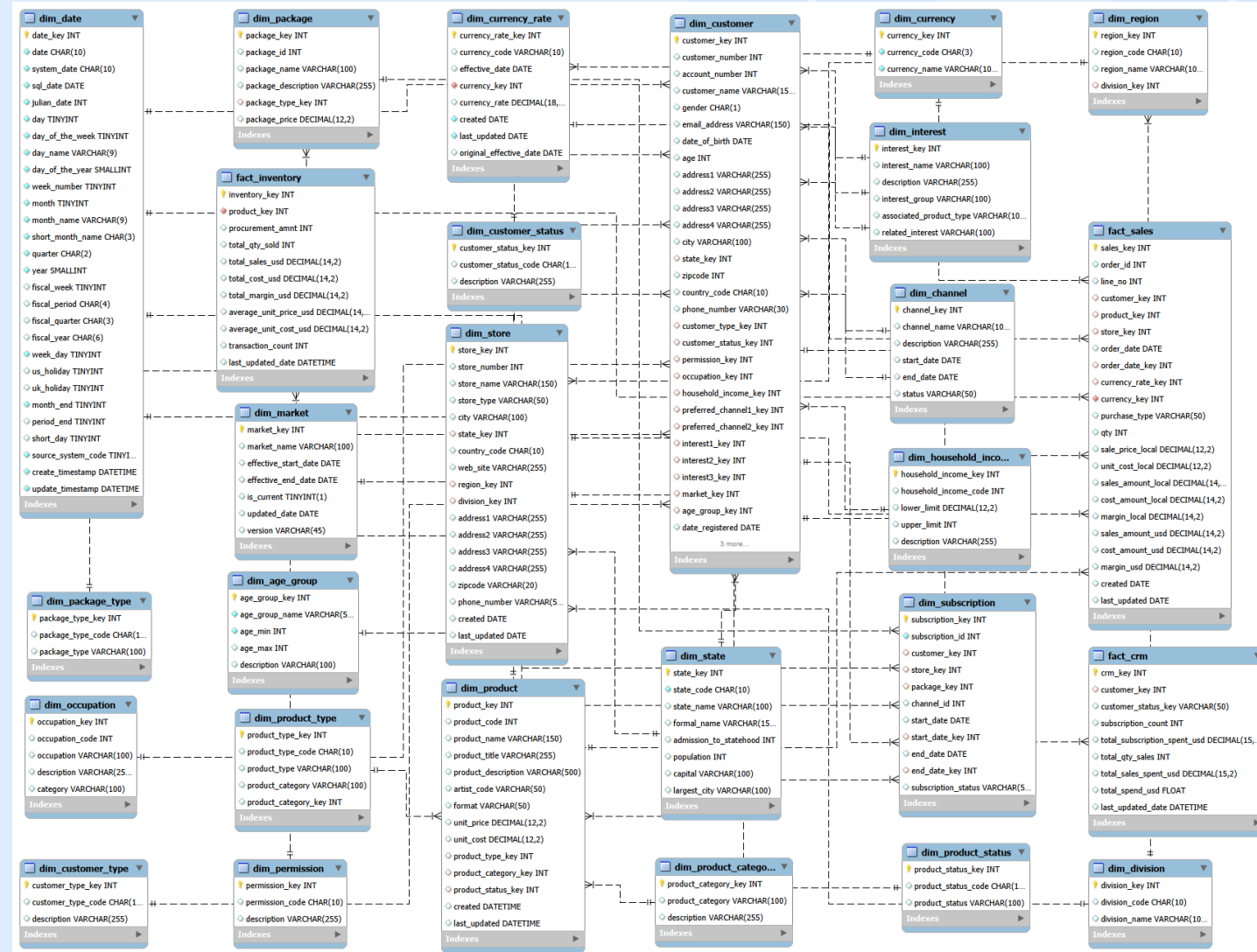


DESIGN DATA WAREHOUSE BY BUSINESS
OPERATIONAL NEEDS

Dimensional Model Design

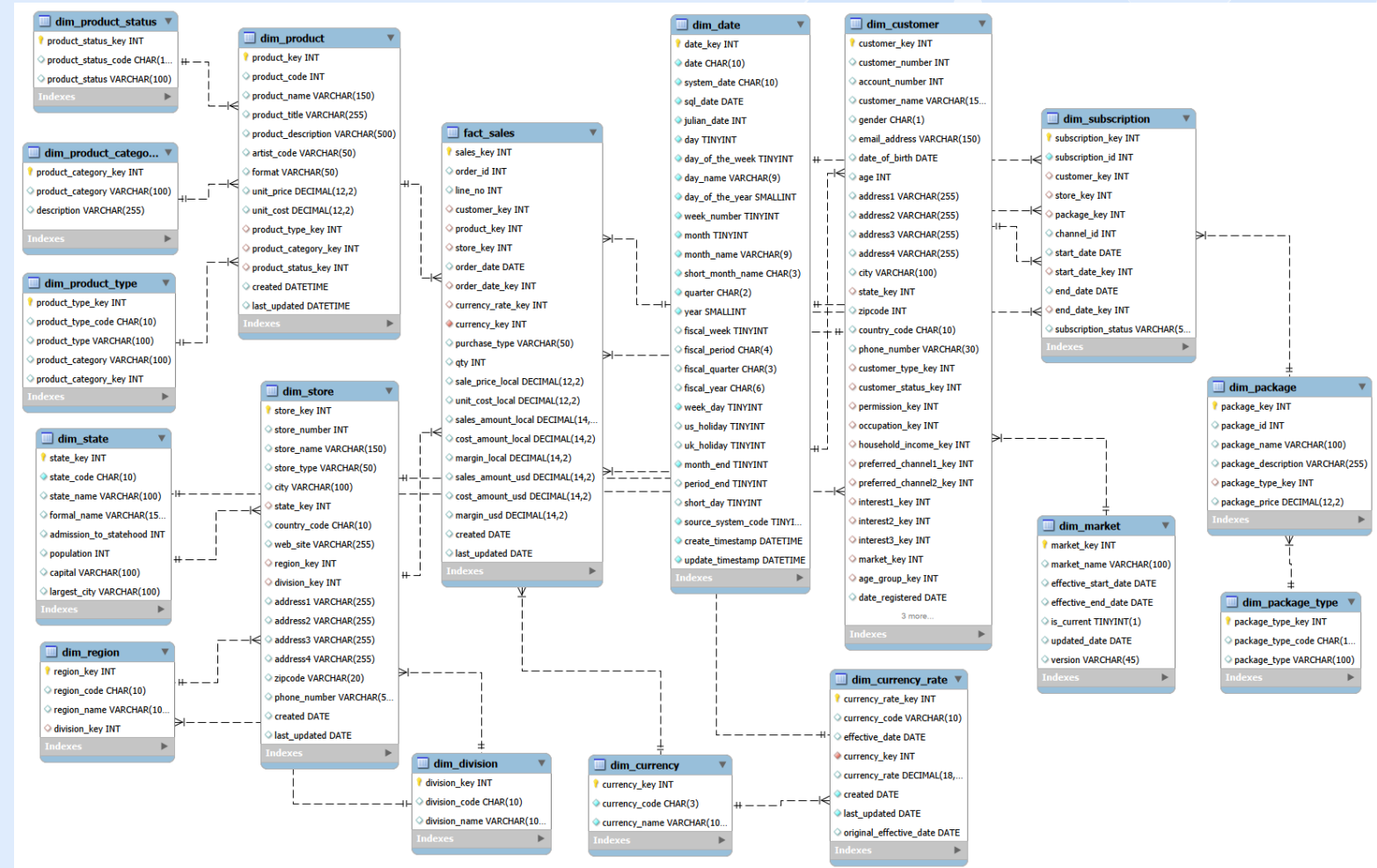
DESIGN DIMENSIONAL MODEL FOR CONSISTENCY

- The dimensional model is designed to create a **single, integrated analytical layer** that combines data from both the Sales system (Informix) and ERP system (Oracle), enabling ICE Entertainment to achieve a unified view of performance across the business.
- The architecture follows a **star schema approach**, centred around three core fact tables: **fact_sales**, **fact_inventory**, and **fact_crm**, each representing a key business operation: sales, purchasing, and customer analytics.
- To ensure consistency across these domains, the model uses **conformed dimensions** including product, store, time, currency, and market. These shared dimensions allow seamless cross-analysis, such as linking customer behavior to product profitability or evaluating performance across geographic hierarchies and time periods. The time dimension is particularly important as it supports both **calendar and fiscal reporting**, as well as seasonal analysis required by the business.
- From a design perspective, each fact table is defined at a clear grain: sales at the **order line level**, inventory as a **periodic snapshot**, and customer as a **historical profile (SCD Type 2)**. Currency conversion is standardised using monthly exchange rates to ensure all reporting is in USD.



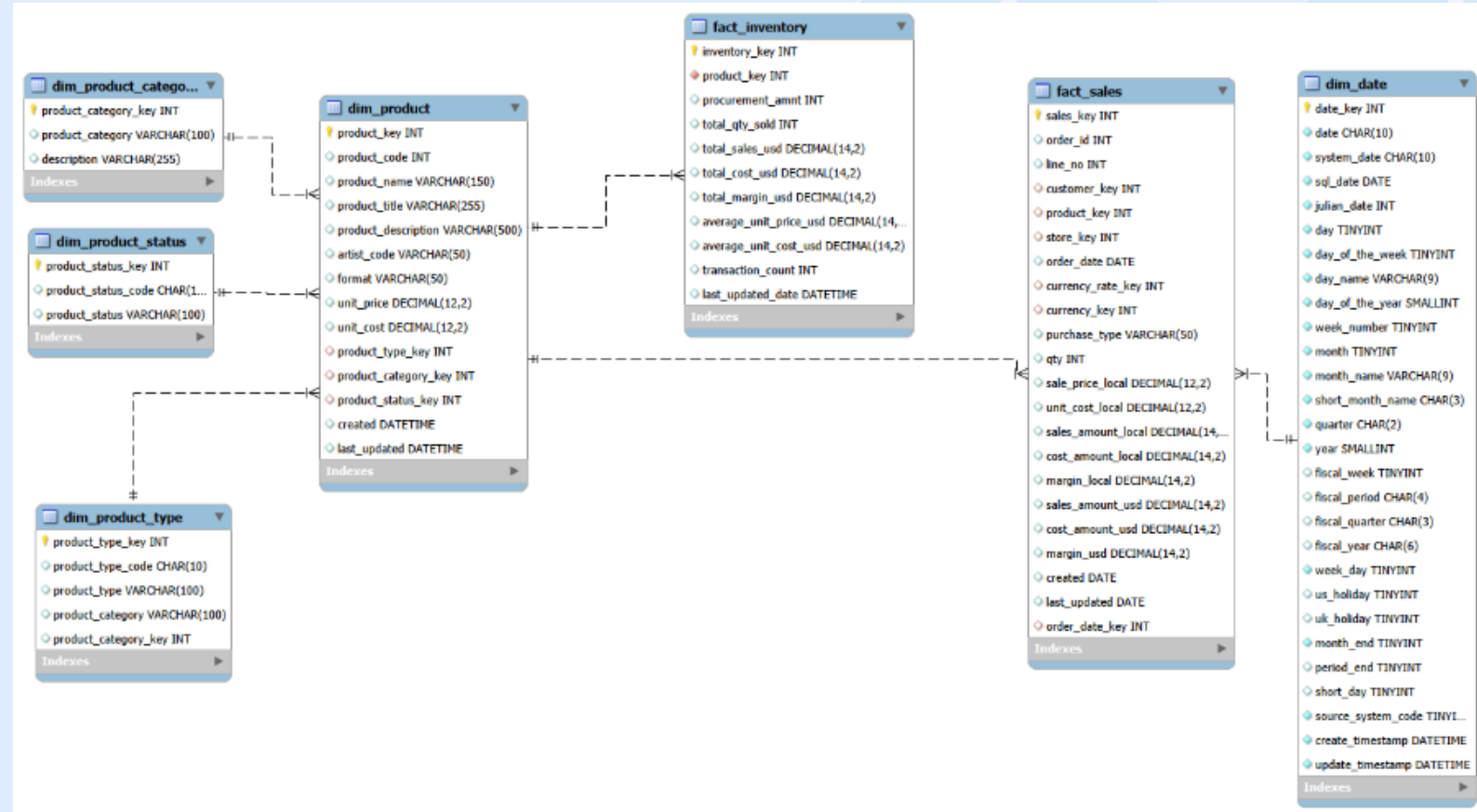
SALES OPERATION IN DIMENSIONAL MODEL (FACT SALES)

- The sales domain is designed to support ICE Entertainment's primary requirement of achieving **global sales visibility and profitability analysis**. The **fact_sales** table captures transactional data at the **order line level**, allowing detailed analysis of revenue, cost, and margin over time across multiple dimensions including product, customer, store, and currency.
- This fact table integrates data from both source systems: sales transactions from the Sales system and product cost data from the ERP system. This integration enables accurate calculation of key business metrics, where sales are derived from quantity and price, cost is based on the latest unit cost prior to sale, and margin is calculated as the difference between the two. Importantly, all values are converted into USD using the currency dimension, ensuring consistency in global reporting.
- The design supports multi-level analysis, including customer profitability, product performance by category and season, and store performance across geographic hierarchies. By structuring the model in this way, decision-makers can drill down from high-level summaries to detailed transaction-level insights, enabling more data-driven decisions across the organisation.



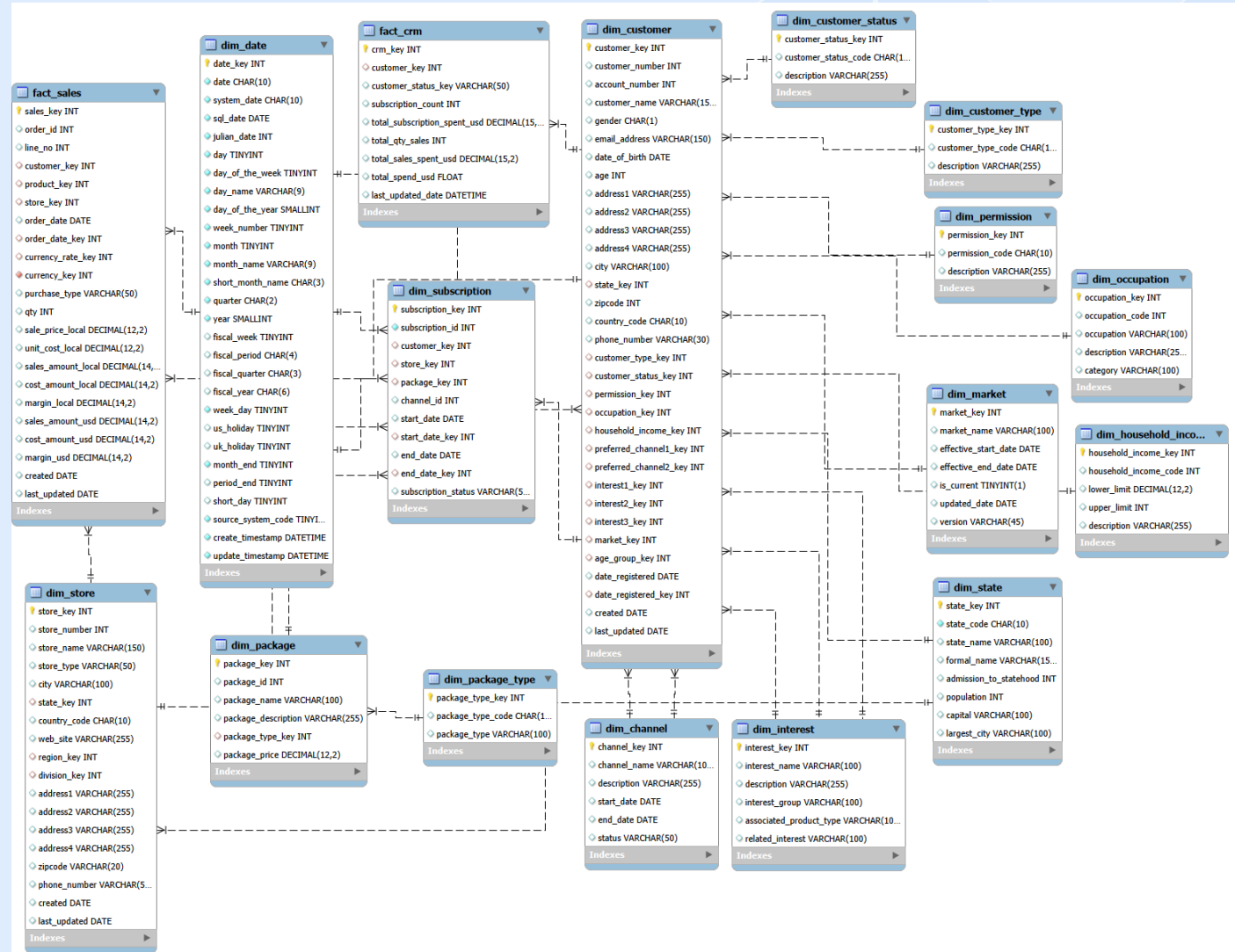
PROCUREMENT OPERATION IN DIMENSIONAL MODEL (FACT INVENTORY)

- The purchasing domain focuses on enabling **supplier performance evaluation and inventory intelligence**, addressing the current lack of visibility into procurement effectiveness. The **fact_inventory** table is designed at a **snapshot level**, capturing product stock levels and cost information across time, store, and market.
- This table includes key measures such as procurement, received, and closing stock quantities, as well as reorder levels and unit cost. It is supported by dimensions that allow analysis of inventory trends and procurement decisions across different categories.
- A key design consideration is the nature of ICE Entertainment's bulk purchasing model, where costs are allocated across products from large procurement deals. The model also accounts for external streaming costs, which introduce additional variability in cost structures. By linking inventory data with sales data, the model enables deeper analysis of **product profitability and investment efficiency**, helping identify which categories or suppliers deliver the most value.



CUSTOMER ANALYTICS IN DIMENSIONAL MODEL (FACT CUSTOMER)

- The customer domain provides the foundation for ICE Entertainment's future customer loyalty program and behavioral analytics. Instead of a traditional dimension, the model uses a **dim_customer** table structured as a Slowly Changing Dimension (Type 2), allowing the system to track changes in customer attributes over time.
- The fact table is an Accumulating Snapshot Fact Table – **fact_crm**, which stores aggregated customer performance data by unique customer_key, links to **dim_customer** for customer profile data.



Technical Implementation



FROM WAREHOUSE DESIGN TO MANAGEMENT INSIGHTS

Helps managers answer the five required business questions and go further into trends, regional comparisons and customer behaviour.

1. Build

Create the warehouse structure

2. Query

Use SQL and reporting tools to analyse performance

3. Decide

Turn insights into actions

Warehouse structure

fact_sales

Measures

quantity
sales_amount_
usd
margin_usd

dim_customer

dim_product

dim_store

dim_date

dim_market

dim_currency

Sample SQL logic

```
1 SELECT m.market_name,  
2       SUM(f.margin_usd)  
AS total_margin  
3 FROM fact_sales f  
4 JOIN dim_market m ON  
f.market_key = m.market_key  
5 GROUP BY m.market_name  
6 ORDER BY total_margin  
DESC;
```

What management gets

- answers the five required business questions
- compares stores, regions and customer segments
- reveals trends across months, quarters and fiscal periods

The background is a solid blue color with a repeating pattern of overlapping circles and leaf-like shapes. The circles are arranged in a grid, and the leaf-like shapes are positioned at the intersections of the circles, creating a decorative, organic pattern.

Thank You